

FACULTY OF ENGINEERING AND BASIC SCIENCES**ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE****COURSE: ETL (G01)****LAB-2. From Business Requirements to a Dimensional Data Warehouse****1. Context**

In this laboratory, the focus moves to **dimensional modeling and Data Warehouse design**, so that the data can be organized to answer business questions efficiently and consistently.

The objective is not to build a complex ETL pipeline. The provided sources are intentionally simple and mostly clean. Your main challenge is to translate business requirements into a dimensional model, implement the model, load it correctly, and demonstrate that it supports the required analytical outputs.

2. Learning Objectives

- Translate business requirements into dimensional modeling decisions.
- Apply the four-step dimensional design process: business process, grain, dimensions, and facts.
- Design and implement a Star Schema using surrogate keys and referential integrity.
- Load dimension tables and a fact table in the correct sequence.
- Validate the dimensional model through SQL queries and KPIs linked to business requirements.
- Document the system architecture and design decisions in a reproducible GitHub repository.

3. Business Scenario

A retail technology company operates two physical stores and one national online store. Management wants to consolidate six months of sales information into a Data Warehouse to support recurring analytical queries and future dashboards.

The analytical solution must be designed around the business questions below. The model should contain the data necessary to answer them, without simply copying every source field into the Data Warehouse.

4. Business Requirements

ID	Business Requirement
R1	Monitor monthly net sales trends and identify periods of growth or decline.
R2	Compare sales performance across stores and sales channels over time.
R3	Identify the top-performing product categories and brands using revenue and units sold.
R4	Evaluate promotion performance by comparing sales, units, and discounts across promotion types.

R5	Analyze gross profit and gross margin by product category, store, and month.
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Important: Every dimension, fact, measure, KPI, query, and visualization must be justified by at least one business requirement.

5. Available Data Sources

You will receive two input sources. These files are already suitable for the focus of this laboratory; only minimal preparation is required before loading the dimensional model.

Source 1 — sales_transactions.csv

Field	Description
sale_line_id	Unique identifier of each sales line.
transaction_id	Identifier of the sales transaction.
sale_date	Date of the transaction.
store_id	Store identifier.
product_id	Product identifier.
channel_id	Sales channel identifier.
promotion_id	Promotion identifier.
quantity	Units sold.
unit_price_sale	Unit price actually paid by the customer (COP).

Source 2 — reference_data.json

The JSON file contains four reference collections:

- products: product_id, product_name, category, brand, list_price, unit_cost.
- stores: store_id, store_name, city, region, channel_id.
- channels: channel_id, channel_name.
- promotions: promotion_id, promotion_name, discount_pct.

Dataset coverage: 1,000 sales lines, January–June 2026, two physical stores, one online store, four product categories, four brands, and multiple promotion conditions.

6. Expected System

Before writing code, represent the system as connected blocks. A possible conceptual flow is: **Business Requirements** → **Source Data** → **ETL** → **Dimensional Model** → **SQL/KPIs** → **Business Decisions**

Systems Thinking: For each block, identify its inputs, responsibility, outputs, and connection with the rest of the system.

7. Activities

Part A — Requirements Traceability

Review the five business requirements and complete a traceability table before designing the schema.

Requirement	Analytical Question	Required Data	Expected Query	KPI /
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R1			
R2			
R3			
R4			
R5			

Guiding question: What information must the Data Warehouse contain in order to answer each requirement?

Part B — Dimensional Model Design

Apply the four-step dimensional design process in this order:

- Step 1 — Select the business process.
- Step 2 — Declare the grain of the fact table.
- Step 3 — Identify the dimensions required by the business questions.
- Step 4 — Identify the facts/measures required by the business questions.

Your design must explicitly state:

- Business process.
- Fact table grain in one clear sentence.
- Dimension tables and their main attributes.
- Fact table measures and how they are calculated.
- Primary keys, surrogate keys, and foreign-key relationships.

Design rule: The model must be driven by the five business requirements, not by the number of columns available in the source files.

Part C — Star Schema Diagram

Create a Star Schema diagram using diagrams.net, Mermaid, PowerPoint, or another suitable tool. The diagram must show:

- Fact table.
- Dimension tables.
- Primary keys.
- Foreign keys.
- Fact table measures.
- Relationships between tables.

Part D — Data Warehouse Implementation

Implement the dimensional model in SQLite. Create the dimension tables and fact table using SQL or Python-generated SQL.

- Use surrogate integer keys for dimensions.
- Define primary-key and foreign-key relationships.
- Use meaningful data types.
- Preserve the declared grain of the fact table.
- Do not create additional dimensions unless they are justified by a business requirement.

Part E — Minimal Load Pipeline

Implement a simple load process. The purpose of this part is to populate the model, not to build a complex ETL solution.

- Read sales_transactions.csv and reference_data.json.
- Create the Date dimension from sale_date.
- Load dimensions first and generate surrogate keys.
- Map source identifiers to dimension surrogate keys.
- Calculate only the measures required by the business requirements.
- Load FactSales after all required dimension keys are available.

Suggested analytical measures:

Measure	Suggested Calculation
Quantity	Source quantity
Gross Sales	quantity × list_price
Net Sales	quantity × unit_price_sale
Discount Amount	Gross Sales – Net Sales
Cost Amount	quantity × unit_cost
Gross Profit	Net Sales – Cost Amount
Gross Margin %	Gross Profit / Net Sales × 100 (calculate in queries/KPIs)

Important: Do not store derived percentages such as gross margin in the fact table unless you can justify the aggregation behavior. Prefer to calculate them at query time.

Part F — Validate the Model Against the Requirements

Implement at least one SQL query for each business requirement. The purpose is to demonstrate that the dimensional model actually supports the business needs.

Requirement	Minimum Analytical Evidence
R1	Monthly net sales trend.
R2	Sales by store and by channel, optionally segmented by month.
R3	Sales and units by category and brand.
R4	Sales, units, and discount amount by promotion.
R5	Gross profit and gross margin by category/store/month.

Part G — Simple Analytical Outputs

Create only two visualizations. The objective is to validate that the Data Warehouse supports analysis; visualization design is not the main focus of this laboratory.

- Visualization 1: choose the most useful chart for one temporal requirement.
- Visualization 2: choose the most useful chart for one comparative requirement.

Charts may be generated with Matplotlib, Power BI, or another tool, but they must use data queried from the Data Warehouse, not directly from the raw source files.

8. Recommended Repository Structure

lab2-dimensional-dw/

```

|
|— data/
|   |— sales_transactions.csv
|   |— reference_data.json

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|
|— src/
|   |— create_schema.py
|   |— load_dimensions.py
|   |— load_fact.py
|   |— queries.py
|   |— main.py
|
|— database/
|   |— retail_dw.db
|
|— docs/
|   |— star_schema.png
|
|— README.md
|— requirements.txt
|— .gitignore

```

9. Deliverable

Submit only the GitHub repository URL through the LMS. The repository must be complete, reproducible, and contain all files required to review the work.

The README.md must include:

- Project objective and business scenario.
- Five business requirements.
- System architecture / pipeline diagram.
- Business process and declared fact-table grain.
- Star Schema diagram and design justification.
- Description of dimensions, facts, and measures.
- Load order and surrogate-key strategy.
- Execution instructions.
- SQL queries/KPIs mapped to the business requirements.
- Two analytical visualizations and a brief interpretation of the results.

10. Assessment Rubric

Criterion	Evidence	Weight
Requirements Traceability	The five business requirements clearly drive the model, measures, queries, and outputs.	15%
Dimensional Model Design	Correct business process, grain, dimensions, facts, Star Schema, and justified relationships.	30%
Data Warehouse Implementation	Correct schema implementation, surrogate	20%

	keys, PK/FK relationships, and grain preservation.	
Load Process	Dimensions loaded before fact table; source identifiers correctly mapped to surrogate keys; required measures calculated.	15%
Analytical Validation	Five requirement-based SQL queries/KPIs and two meaningful analytical visualizations.	10%
Repository & Documentation	Clear structure, reproducibility, README, architecture, and execution instructions.	10%

11. Final Reflection

Answer briefly in the README:

- How did the business requirements influence your dimensional model?
- What would be the impact of choosing an incorrect grain?
- Did your final model contain any table or attribute that was not necessary for the selected requirements? If so, why?

Think in Systems. Build Data Solutions. Create Value.